Jeonghyun Byun, Ph.D.

Postdoctoral Researcher | Fulbrighter

Kumar Robotics Lab, GRASP, University of Pennsylvania, PA, USA

quswjdgus97@gmail.com | jh-byun.github.io

Education

Seoul National University — *Seoul, Republic of Korea*

Ph.D. in Aerospace Engineering — *March, 2020 - February, 2025*

- Dissertation: Aerial physical interaction strategy considering changes in dynamics
- Advisor: Prof. H. Jin Kim

Seoul National University — *Seoul, Republic of Korea*

B.S. in Aerospace Engineering — *March, 2016 - February, 2020*

- Graduated **top of the department**
- Awarded Summa Cum Laude (GPA: 4.05/4.3)

Professional Experience

University of Pennsylvania — *Philadelphia, Pennsylvania, USA*

Visiting Scholar (Postdoc) — *August, 2026 -*

- **Partially Funded by Fulbright** (August, 2026 - July, 2027)
- P.I.: Prof. Vijay Kumar

Institute of Control, Robotics and Systems (ICROS) — *Seoul, Republic of Korea*

Editorial Board Member — *January, 2026 -*

Editorial Board Member of the Institute of Control, Robotics and Systems (ICROS)

Automation and Systems Research Institute (ASRI), Seoul National University — *Seoul, Republic of Korea*

Postdoctoral Researcher — *March, 2025 - July, 2026*

- **Alternative military service** as technical research personnel (March 2025 - February 2026)
- P.I.: Prof. H. Jin Kim

Publications

Manuscripts Under Review or In Preparation

Jeonghyun Byun[1], Yeonjoon Kim, Jaewoo Lee, and H. Jin Kim "Task-Agnostic Safe and Robust Controller for Compliant Aerial Manipulation: Hybrid Systems Approach"
In Preparation, **IEEE Transactions on Robotics**

Lin Yang[1], Jinwoo Lee, Domenico Campolo, H. Jin Kim, **Jeonghyun Byun**[†] "Whole-body motion planning and safety-critical control for aerial manipulation."
Under review, **Mechatronics**

Jeonghyun Byun[1], Dohyun Eom, Dongjae Lee, H. Jin Kim. "Motion/force control against discontinuous contact and friction forces for aerial push-and-slide operation."
Conditionally Accepted, **IEEE Transactions on Control Systems Technology**

Journals Articles

Yongjae Lim[1], Youngmin Yoon, **Jeonghyun Byun**, Sangyoon Kim, and H. Jin Kim. "Reachable-Set-based Trajectory Sampling for Local Planning of Autonomous Vehicles." **IEEE Access**, 2025

Yeongin Song[1], Hyunmin Kim[1], **Jeonghyun Byun**, Keun Park, Murim Kim, and Seung Jae Lee. "Aerial Dockable Multirotor UAVs: Design, Control and Flight Time Extension through In-flight Battery Replacement." **IEEE Access**, 2025

Jeonghyun Byun[1], Junha Kim, Dohyun Eom, Dongjae Lee, Changhyeon Kim, H. Jin Kim. "Image Based Time-Varying Contact Force Control of Aerial Manipulator using Robust Impedance Filter." IEEE Robotics and Automation Letters

(**RA-L**), 2024. *(*Orally presented at IROS 2024 held in Abu Dhabi, UAE.)*

Jeonghyun Byun[1], Inkyu Jang, Dongjae Lee, H. Jin Kim. "A Hybrid Controller Enhancing Transient Performance for an Aerial Manipulator Extracting a Wedged Object." IEEE Transactions on Automation Science and Engineering (**T-ASE**), 2023. *(*Orally presented at ICRA 2024 held in Yokohama, Japan.)*

Dongjae Lee[1] **Jeonghyun Byun**, H. Jin Kim. "RISE-based trajectory tracking control of an aerial manipulator under uncertainty."
IEEE Control Systems Letters (**L-CSS**), 2022.

Peer-Reviewed Conferences

Yubin Kim[1], Jinwoo Lee, Yongjun You, H. Jin Kim, **Jeonghyun Byun**[†]. "Safety-Critical Bilateral Teleoperation for Omnidirectional Aerial Manipulation Using Force-Sensorless Haptic Feedback." 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**).

Jeonghyun Byun[1], Yeonjoon Kim, Dongjae Lee, H. Jin Kim. "Safety-Critical Control for Aerial Physical Interaction in Uncertain Environment." 2025 International Conference on Robotics and Automation (**ICRA**).

Jeonghyun Byun[1], Dohyun Eom, H. Jin Kim. "Haptic-Based Bilateral Teleoperation of Aerial Manipulator for Extracting Wedged Object with Compensation of Human Reaction Time." 2024 International Conference on Unmanned Aircraft Systems (**ICUAS**).

Dongjae Lee[1], Sunwoo Hwang, **Jeonghyun Byun**, H. Jin Kim. "Autonomous Aerial Perching and Unperching Using Omnidirectional Tiltrotor and Switching Controller." 2024 International Conference on Robotics and Automation (**ICRA**).

Inkyu Jang[1], Sunwoo Hwang, **Jeonghyun Byun**, H. Jin Kim. "Safe Receding Horizon Motion Planning with Infinitesimal Update Interval." 2024 International Conference on Robotics and Automation (**ICRA**).

Jeonghyun Byun[1], Byeongjun Kim, Changhyeon Kim, Donggeon David Oh, H. Jin Kim. "Stable Contact Guaranteeing Motion/Force Control for an Aerial Manipulator on an Arbitrarily Tilted Surface." 2023 International Conference on Robotics and Automation (**ICRA**).

Byeongjun Kim[1], Dongjae Lee, **Jeonghyun Byun**, H. Jin Kim. "Globally Defined Dynamic Modelling and Geometric Tracking Controller Design for Aerial Manipulator." 2023 International Conference on Robotics and Automation (**ICRA**).

Dongjae Lee[1], Inkyu Jang[1], **Jeonghyun Byun**, Hoseong Seo, H. Jin Kim. "Real-Time Motion Planning of a Hydraulic Excavator using Trajectory Optimization and Model Predictive Control." 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**).

Jeonghyun Byun[1], Dongjae Lee, Hoseong Seo, Inkyu Jang, Jeongjun Choi, H. Jin Kim. "Stability and Robustness Analysis of Plug-Pulling using an Aerial Manipulator." 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**).

Projects

Autonomous Wheel Loader *South Korea*
Hyundai Construction Equipment (HCE) *March, 2023 - Present*
(Leading the team of graduate students) Develop trajectory generation strategy for V-shape maneuver and loading/unloading of a wheel loader,

Research and Education on Defense Intelligent Swarm System *South Korea*
Ministry of Science and ICT *July, 2024 - December, 2025*
(Led the team of graduate students) Proceed on control and planning of single and multiple unmanned aerial vehicles

Hybrid Motion/Force Controller for Underactuated Aerial Manipulator *South Korea*
BrainKorea21PLUS *December, 2021 - March, 2022*
(Independent Research Project) Design a transient performance-enhancing hybrid motion/force controller for an underactuated multirotor equipped with added equipment

Friction Coefficient Estimation *South Korea*
Hyundai Motors *June, 2021 - May, 2022*
Physically estimate friction coefficient between car's tire and road

Autonomous Excavator *South Korea*
Hyundai Construction Equipment (HCE) *September, 2020 - January, 2021*
Design external wrench estimator for excavator path-planning

Invited Presentations

2026 **Task-Agnostic Controller for Robust and Safe Aerial Manipulation: Hybrid Systems Approach**, Hanyang University ERICA, South Korea

2026 **Task-Agnostic Controller for Robust and Safe Aerial Manipulation: Hybrid Systems Approach**, Glocal Lab, Department of Robotics and Mechatronics Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), South Korea

2025 **Aerial Physical Interaction Strategy Considering Changes in Dynamics**, Control and Dynamic Systems Lab, Sookmyung Women's University, South Korea

2025 **Aerial Physical Interaction Strategy Considering Changes in Dynamics**, Workshop on Advanced Aerial Robotics at Dragon Lab, University of Tokyo, Japan

2025 **Safety-Critical Aerial Physical Interaction**, Advancements in Aerial Physical Interaction (Workshop) in IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), China

2025 **Introduction to Drone Research in LARR (Korean)**, Seoul National University, South Korea

Honors

Awards

July, 2026 **1st Place, 2026 MathWorks Special Paper Award**, Institute of Control, Robotics and Systems (**ICROS**)

Feb., 2026 **Promotional Prize Winner in 2025 Best Video Award**, **IEEE Access**

July, 2025 **Selected Paper for Journal Submission - 40th ICROS Annual Conference**, Institute of Control, Robotics and Systems (**ICROS**)

Nov., 2022 **3rd Place, Aerospace Paper Award**, Korea Aerospace Industries (**KAI**), LTD.

Feb., 2020 **Top Graduate Award**, Department of Aerospace Engineering, Seoul National University

Feb., 2020 **Summa Cum Laude**, Department of Aerospace Engineering, Seoul National University

Sep., 2018 **Special Recognition**, 7th SNU Creative Design Fair, College of Engineering, Seoul National University

Sep., 2017 **Special Recognition**, 6th SNU Creative Design Fair, College of Engineering, Seoul National University

Fellowships

Aug., 2026 **Fulbright Visiting Scholar Program (Postdoctoral Researcher)**, Fulbright Korea

Nov., 2021 **BK21 Excellent Research Talent Fellowship**, BrainKorea21PLUS

Mar., 2020 **BK21 PLUS Doctoral Fellowship**, BrainKorea21PLUS

Skills

Programming: C/C++, Python, ROS, ROS2, MATLAB/Simulink, Arduino

Language: Korean (native), English (proficient), French (elementary)

Tools: Git, CAD (Solidworks, Fusion360, Onshape), Optimization Toolbox/Solver (Acados, CasADi, CPLEX)

Academic Services

- **Journal reviewer for IJRR**, 2025
- **Journal reviewer for IEEE T-RO**, 2021, 2024, 2026

- **Journal reviewer for IEEE T-ASE**, 2023 - 2024
- **Journal reviewer for IEEE T-IE**, 2025
- **Journal reviewer for IEEE RA-L**, 2022 - 2026
- **Journal reviewer for Springer IJCAS**, 2023
- **Conference reviewer for IEEE ICRA**, 2022-2023, 2025-2026
- **Conference reviewer for IEEE IROS**, 2023, 2025-2026
- **Conference reviewer for IFAC World Congress**, 2026

Teaching Experiences

2026	**Instructor, Intelligent Autonomous Vehicle Operation Practicum, Seoul National University**, Instructed undergraduate and graduate students on compact drone operation and control
2021	**Tutor, Engineering Maths 1, Seoul National University**, Solved several difficult problem sets
2020	**TA, Introductory Engineering Probability, Seoul National University**, Developed scoring criteria for the exams
2020	**Tutor, Physics 2, Seoul National University**, Solved several difficult problem sets
2020	**TA, Engineering Maths 1, Seoul National University**, Developed scoring criteria for the exams
2017.03 - 2018.06	**Tutor, Physics, Seoul National University**, Solved some difficult problem sets

References

Prof. H. Jin Kim (Seoul National University, hjinkim@snu.ac.kr)

Prof. Domenico Campolo (Nanyang Technological University, d.campolo@ntu.edu.sg)

Prof. Seungjae Lee (Seoul National University of Science and Technology, seungjae_lee@seoultech.ac.kr)